LECTURE NOTES

Topic 1: Introduction to Programming Development (1 hour)

Learning Outcomes:

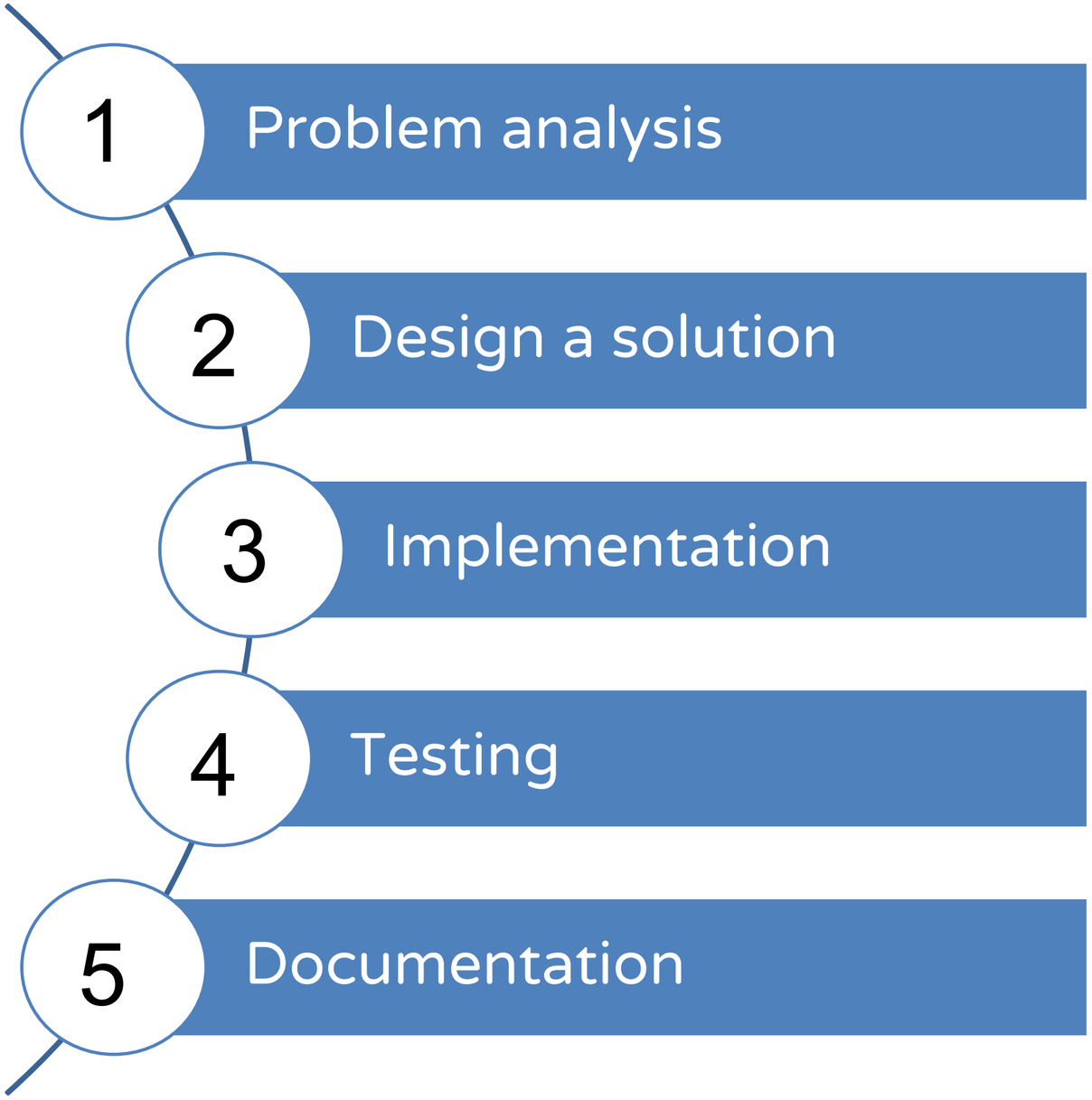
At the end of the lesson, student should be able to:

(a) Describe steps in problem solving/ program development:

- Problem Analysis
- Design A Solution
- Implementation
- Testing
- Documentation

LECTURE NOTES

Five (5) steps in problem solving



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Five (5) Steps in Problem Solving

1. Problem Analysis

- Problem analysis is the act of identifying **input**, **process** and **output (IPO)**.
- Programmers use **IPO model to** organize and summarize the results of the given problem.

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Five (5) Steps in Problem Solving

1. Problem Analysis using IPO model

- IPO model
  Helps break down the problem to understand what is needed.

Input (I)	Process (P)	Output (O)

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Five (5) Steps in Problem Solving

2. Design a Solution

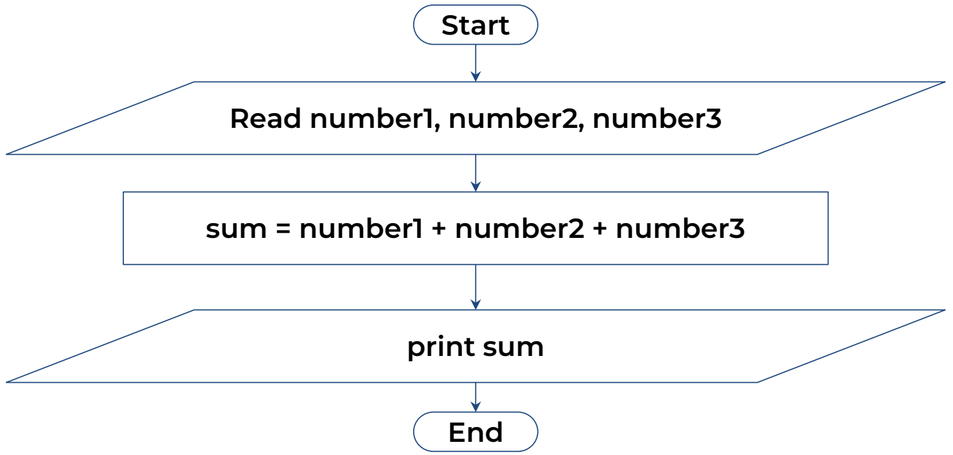
- Design a solution involves creating an **algorithm**
- Algorithm is **logical sequence of steps** to solve a problem
- The logical sequence of steps in algorithm are derived from the input, process and output of the problem analysis step.

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Five (5) Steps in Problem Solving

2. Design a Solution

- There are two (2) types of algorithm representations:
  - **Pseudocode** -Written steps in simple language.
  - **Flowchart** - Visual diagram of the steps.

Pseudocode	Flowchart
Start Read number1, number2, number3 sum = number1 + number2 + number3 Print sum Stop	

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Five (5) Steps in Problem Solving

3. Implementation

- Implementation is applying the algorithm by writing a **computer program** using a programming language

  - ❖ Python
  - ❖ Java
  - ❖ PHP
  - ❖ C++
  - ❖ Kotlin
  - ❖ C#

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Five (5) Steps in Problem Solving

3. Implementation

- Example Python coding

```
# Ask the user to enter three numbers
num1 = float(input("Enter the first number: "))  # Get first number
num2 = float(input("Enter the second number: "))  # Get second number
num3 = float(input("Enter the third number: "))  # Get third number

# Calculate the average
average = (num1 + num2 + num3) / 3

# Display the result
print(f"The average of {num1}, {num2}, and {num3} is: {average:.2f}")  # Display result
```

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Five (5) Steps in Problem Solving

4. Testing

- Ensure the program **works correctly** and fixes errors (bugs).
- **Three types of errors:**
  - ✅ **Syntax Errors** – Incorrect coding grammars
    - Example: missing brackets ( )
  - ✅ **Logic Errors** – Incorrect calculations or wrong program behavior.
    - Example: incorrect average calculation
  - ✅ **Run-time Errors** – Errors that occur while the program is running
    - Example: division by zero

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Five (5) Steps in Problem Solving

5. Documentation

- A detailed **write-up** explaining how the program works.
- Helps future programmers **understand, maintain, or update** the code.
- Typical program **documentation materials** include:
    - the origin and nature of the problem
    - a brief narrative description of the program
    - data-record descriptions
    - program listing
    - testing results